

production. This technology is constantly evolving and improving in the manufacturing industry. Improvements in speed, quality and materials could make 3D printing useful in mass production and mass customisation. The combination of AI and 3D printing brings lots of new excitement and flexibility in additive manufacturing technology. The role of AI in 3D printing technology is in the software. In other words, the intelligence of a 3D printer lies in the software program. AI algorithms are being incorporated into 3D modeling programs to help you optimise the 3D printing process and also create the best 3D printable models.

3. AI and IoT

The emerging technologies like AI and IoT have been going on well especially in the manufacturing industry. These technologies have been revolutionising the mass production of goods and boosting the output of different industries.

In the manufacturing industry, the IoT involves the use of devices/sensors, cloud computing and the Internet for communication, and AI software to compile massive amounts of data for intelligent machine learning. Through IoT technology, machines can seamlessly talk to each other and react to any problems that arise. IoT system can quickly alert other machines and human workers—allowing the problem to be addressed in real-time.

4. AI-robot

When we talk about robots in manufacturing industries, there are two technology terms that come into the picture: cobots and robotic process automation (RPA). Cobots were first coined by professors at Northwestern University in America. Unlike traditional industrial robots, cobots or collaborative robots are the latest generations of robotic systems intended to work alongside humans. While cobots are robots intended to interact with humans in a shared space or to work safely in close proximity, RPA is a form of business process automation technology based on metaphorical software bots. That is, unlike the physical robotic machines, RPA is software that acts as an additional employee in various disciplines throughout the supply chain, automating certain activities, minimising human errors, and maximising productivity. Actually, RPA works behind the scenes, replicating human minds, accelerating operations and transforming business processes.

The robotics market is growing at a great speed across the globe. There are millions of robots working in factories all over the world. With the emergence of AI, robots are becoming smarter, cheaper and easier to use. Cutting-edge AI techniques will help the industrial robots to improve the speed, accuracy and safety of production.

The most common robots are pick and place, packaging, welding and assembly robots. Other manufacturing robots include cutting, drilling, milling and painting robots. Assembly robots are used on automated assembly lines including automotive, consumer electronics and electrical appliances. AI-powered computer vision is one of the most important modern researches in robotics. Some assembly robots employ computer vision and sen-

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